

B.Tech IV Year I Semester (R20) Supplementary Examinations April/May 2025

AGILE METHODOLOGIES
(Computer Science & Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Explain Agile Manifesto and its importance. | 2M |
| (b) Differentiate between Agile Model and Traditional Model. | 2M |
| (c) What is SCRUM, and how does it benefit software development? | 2M |
| (d) Discuss the principles of Extreme Programming (XP). | 2M |
| (e) Explain the role of knowledge management in Agile development. | 2M |
| (f) Describe the Agile Knowledge Sharing Model. | 2M |
| (g) How does Agile address the challenge of managing unstable requirements? | 2M |
| (h) Explain the Agile Requirements Abstraction Model. | 2M |
| (i) Discuss the significance of Test-Driven Development in Agile. | 2M |
| (j) What are the challenges of Agile in global software development? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Describe the Agile Project Management process. Discuss the key roles involved in Agile teams. 10M
- OR**
- 3 Compare and contrast Agile Software Development with Traditional Software Development Models. 10M
- 4 Explain the lifecycle of Feature-Driven Development. How does it ensure continuous improvement? 10M
- OR**
- 5 Discuss the various roles and practices in Extreme Programming (XP). 10M
- 6 Elaborate on the Agile Information Systems and how they support decision-making in organizations. 10M
- OR**
- 7 What are the challenges of migrating to Agile methodologies, and how can they be mitigated? Explain in detail. 10M
- 8 (a) Explain the Agile Requirements Engineering Process. 5M
(b) How does Agile manage requirements prioritization? 5M
- OR**
- 9 Describe the Agile Requirements Modeling and Generation process. 10M
- 10 Discuss the Agile approach to Quality Assurance. How is it different from traditional methods? 10M
- OR**
- 11 Explain Agile Metrics and their importance in Agile Product Development. 10M

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

AGILE METHODOLOGIES
(Computer Science & Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) Define Agile Software Development. | 2M |
| (b) Explain the importance of ethics in Agile Teams. | 2M |
| (c) What are the key characteristics of Crystal methodology? | 2M |
| (d) How does SCRUM differ from other Agile methodologies? | 2M |
| (e) What is the Institutional Knowledge Evolution Cycle in Agile? | 2M |
| (f) Explain the role of Story Cards in Agile Knowledge Sharing. | 2M |
| (g) How is Requirements Management handled in an Agile Environment? | 2M |
| (h) Discuss Agile Requirements Prioritization Techniques. | 2M |
| (i) What are Agile Metrics, and why are they important? | 2M |
| (j) How does Agile ensure Quality Assurance through Test-Driven Development? | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|-----|
| 2 | Describe the Agile Manifesto and discuss how it has influenced modern software development practices. | 10M |
| OR | | |
| 3 | Explain the various Agile Drivers, Capabilities, and Values that support software development. | 10M |
| 4 | Discuss the principles and lifecycle of Adaptive Software Development. | 10M |
| OR | | |
| 5 | Explain the key features of SCRUM and how it contributes to Agile Project Management. | 10M |
| 6 | What are the early schools of Knowledge Management in Agile? Discuss their relevance today. | 10M |
| OR | | |
| 7 | Explain the challenges of Managing Software Knowledge in an Agile environment and provide solutions. | 10M |
| 8 | How does Agile handle Requirements Elicitation? Discuss with examples. | 10M |
| OR | | |
| 9 | Explain Agile Requirements Generation and the importance of concurrency in this process. | 10M |
| 10 | Discuss the role of Financial and Production Metrics in Feature-Driven Development. | 10M |
| OR | | |
| 11 | Explain the challenges and benefits of Agile in Global Software Development. | 10M |

B.Tech IV Year I Semester (R20) Supplementary Examinations July/August 2024

AGILE METHODOLOGIES
(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Differentiate between traditional and agile model for software development. | 2M |
| (b) State the manifesto for agile software development. | 2M |
| (c) What are the principles of lean management? | 2M |
| (d) Why is Crystal known as one of the most flexible Agile methodologies? | 2M |
| (e) Why is it preferable to adopt agile decision-making? | 2M |
| (f) What is institutional knowledge? State an example of institutional knowledge in action. | 2M |
| (g) What are the challenges associated with agile requirements engineering activities? | 2M |
| (h) What is agile requirements gathering? | 2M |
| (i) State the considerations to choose agile metrics. | 2M |
| (j) Briefly explain the Test-Driven Development process. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the following terms with reference to agile software development model: 10M
- (i) Pair programming
- (ii) Refactoring
- OR**
- 3 Enlist the most popular Agile methodologies and elaborate any two of them with their advantages and disadvantages. 10M
- 4 What is Extreme Programming (XP) model? How does it emphasizes speed and continuous delivery? 10M
- OR**
- 5 What is Sprint? Explain the Sprint with its significance in the Scrum lifecycle. 10M
- 6 (a) Does agile software development support knowledge management initiatives? 5M
- (b) Can knowledge management improve team cohesion in agile software development? Justify your answer. 5M
- OR**
- 7 What are Key features of the Story card Maturity Model (SMM) process? How can story cards analysis be done based on SMM value principles? 10M
- 8 How to manage requirements in an agile project management environment? Illustrate the best practices for managing requirements in agile projects. 10M
- OR**
- 9 Enlist agile requirements models. Explain any one of them in detail and differentiate it from the traditional requirements engineering model. 10M
- 10 What are the best practices of agile quality assurance process? Discuss in detail. 10M
- OR**
- 11 Explain the principles of agile product development? Also describe the agile approach to Quality Assurance. 10M

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

AGILE METHODOLOGIES
(Computer Science & Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Briefly explain the theory of Just-in-time inventory. | 2M |
| (b) With reference to agile software development model explain the term “short release”. | 2M |
| (c) What are the fundamental practices on which the Kanban agile methodology relies on? | 2M |
| (d) Prove that the statement “Scrum excels at managing iterative and incremental projects” is true. | 2M |
| (e) What is agile decision-making? | 2M |
| (f) Signify the importance of institutional knowledge. | 2M |
| (g) What is the role of requirements engineering in agile development? | 2M |
| (h) What are the characteristics of concurrent development? | 2M |
| (i) What are agile metrics? | 2M |
| (j) What is agile quality assurance? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Justify the statement “test-driven development, pair programming and refactoring ease code development processes”. 10M
- OR**
- 3 State and explain the principles of agile methodology with their pros and cons as compared to traditional software development methodology. 10M
- 4 Discuss the framework for product development using Feature-Driven Development. 10M
- OR**
- 5 State the components of Extreme Programming (XP) model and its role in the XP lifecycle. 10M
- 6 Enlist and elaborate steps to for a team to be successful in agile decision-making during a project. 10M
- OR**
- 7 Discuss the knowledge life cycle and strategies in each stages of the cycle. 10M
- 8 Discuss the Agile Requirements Abstraction Model in detail. 10M
- OR**
- 9 What is requirements prioritization? How to prioritize requirements in agile? 10M
- 10 What is test-driven development? Should agile software developers use test-driven development? Justify your answer. 10M
- OR**
- 11 Which are the key performance indicator metrics in feature driven development? Explain in detail. 10M
